12.7 Solving One-Step Equations in One Variable Using Addition and Subtraction - Part 1

Lesson Introduction

Benchmark	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction where all terms and solutions are integers.	Learning Target	• I can solve addition and subtraction one-step equations in one variable using manipulatives, drawings, and number lines to show the inverse operations.

	Building On	Working Toward
Benchmark Coherence	MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Essential Questions	• How would you use inverse operations to solve an algebraic equation? • How would models help you solve an algebraic equation using inverse operations?
Lesson Narrative	In this lesson, students solve equations in a different way than they did in previous lessons. Students are reasoning about changes one could make to visual models while keeping the models balanced alongside an equation. Thus, students are thinking about doing things on each side of an equation rather than simply thinking "What value would make this equation true?" Focus on the concept of inverse operations being used to "undo" a problem. Continue working with students to develop their understanding of the vocabulary of inverse operations and properties of equality.

Component Summary	12.7.1 Warm-Up (~5 min.)	Are They Inverses? *(SE p. 313)*	12.7.3 Guided Practice (10 min.)	Practicing solving equations with inverse operations *(SE p. 316)*
	12.7.2 Guided Instruction (20 min.)	Modeling solving equations with inverse operations *(SE p. 314)*	12.7.4 Your Turn! (10 min.)	Using diagrams and inverse operations to solve equations *(SE p. 317)*
	12.7.5 Wrap-Up (~5 min.)	Self-reflecting on the learning of the lesson *(SE p. 319)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: • Addition Property of Equality • Subtraction Property of Equality <u>Additional Terms:</u> • Inverse • Inverse operations	• (Optional) Hanger with objects to hang from it • (Optional) Balance with weights	• N/A

Teacher Tips

Common Misconceptions	Things to Consider
• Students may struggle to understand that what happens on one side of an equation must also occur on the other side in order to keep the equation balanced. • Students may think the mathematical concepts of “inverse” and “opposite” are interchangeable. • Students may struggle with integer operations.	• Emphasize the visual of a balance beam or a hanger to help students understand and remember that meaning of the equals sign and that an operation that is done to one side of the equals sign of an equation must also be done on the other side of the equal sign. • Highlight opportunities to clarify the distinction between “inverse operations” and “opposite values.” For example, in question 3 in 12.7.2 Guided Instruction, for $x + 5 = 11$, the “$+5$” represents both the inverse and opposite. • Observe students who struggle with integer operations to ensure they understand the additive inverse of a negative number. Some students may subtract a negative where adding a negative is the appropriate operation. Consider how small group instruction may be helpful to these students. Employ algebra tiles or other manipulatives to help students build their understanding. • Some students will find that the models are not necessary and they can do the mathematics quickly. Consider how to challenge these students. Consider having these students share their explanations. These students may understand the arithmetic but may not be able to engage in algebraic reasoning.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Notice and Wonder 12.7.2 Guided Instruction and 12.7.3 Guided Practice position students to notice how inverse operations can be used to solve equations. Consider applying Notice and Wonder routines to each visual representation of balancing an equation, positioning students to discover the relationship between doing the same thing on each side of an equation and using inverse operations.	**MA.K12.MTR.6.1: Reasonableness** Students will be working to use inverse operations to solve one-step equations involving addition or subtraction. Clarifications within this standard emphasize the appropriateness of a solution to a problem. Encourage students to substitute their solution back into a problem and to verify their solution by explaining the method used.

Differentiate Instruction

Explain that the bar models, balance, hanger, and number line are all different ways of modeling the same concept of balancing an equation. Consider allowing students to use whichever method makes the most sense to them, like an actual hanger or balance to model the concept of balancing an equation by adding to or removing objects from each side of it. Allow students to use visual aids to solve equations for as long as they need to understand the concept. Encourage students to verbally explain the process they use to solve the equations.

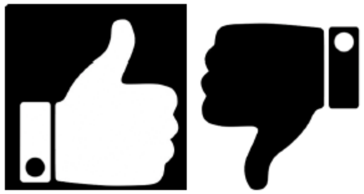
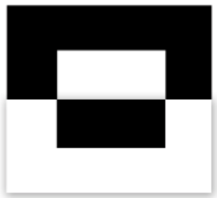
Lesson Notes:

12.7.1 Warm-Up: Are They Inverses?

Component Overview: *Students explore the idea of inverses by examining two black-and-white images. After students explain whether or not they think the images are inverses, further explore the concept by validating students' responses and discussing their working definitions of inverse.*

12.7.1 Warm-Up: Are They Inverses?

Angelica and Felipe were working on a project in art class. They were given the images below and instructed to explain whether or not the images are inverses.

Image 1	Image 2
	

Angelica said, "This was just like inverse operations from math class. Inverse means to do the reverse."
Felipe responded, "But the second image is a smaller image inside of the same image, so it is not a picture of inverses."

1. Explain whether you agree with Angelica, Felipe, or both.

   Answers will vary. I agree with Angelica because the inverse means to reverse the operation. I agree with Felipe because the second image is two different sized rectangles. I agree with both because Angelica has the correct definition and Felipe explained why the shapes in the second image are not inverses.

2. How do you see inverses represented in either image?

   Answers will vary. I see inverses represented in image 1 because the hands are facing opposite directions. I see inverses represented in image 2 because the opposite colors are also on opposite sides.

Encourage students to brainstorm all the examples they can think of in math and in real life that are inverses. Possible examples include addition/subtraction; multiplication/division; the faster you walk, the less time it takes to get to a destination; driving forward and reversing; etc.

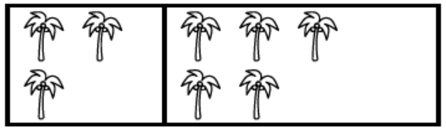
12.7.2 Guided Instruction: Solving Equations With Inverse Operations

Component Overview: *Students will begin exploring how to use inverse operations to solve one-step equations involving addition or subtraction. Students will use bar models and a balance scale to explore doing the same thing on each side of an equation in order to balance the equation.*

12.7.2 Guided Instruction: Solving Equations with Inverse Operations

Bar models and balance scales are useful tools for representing equations.

1. In the bar model shown, each palm tree represents 1 unit.



The bar model can be used to represent either an addition or subtraction equation. Fill in the missing values in each equation to match the bar model.

$3 + \underline{\ 5\ } = 8$

$8 - 3 = \underline{\ 5\ }$

After students create the equations, have them discuss how the equations relate to each other. Guide students toward the idea of addition and subtraction being inverse operations and what that means.

2. The bar model shown can be used to represent either an addition or subtraction equation.

7	
x	4

a. Circle the addition equation that matches the diagram.

$x + 7 = 4$ $\quad$ $7 = x + 4$ $\quad$ $x = 7 + 4$

b. Write an equivalent equation using the inverse operation.

Answers will vary. $x = 7 - 4$; $4 = 7 - x$

c. Justify to your partner that the equation you wrote is an equivalent equation.

Answers will vary. Since I used inverse operations to rewrite the equation, my equation is an equivalent equation because the value of the expressions on both sides of the equation remain equivalent.

To provide additional scaffolding for students who may be struggling to make connections and identify the corresponding equations, ask questions like:

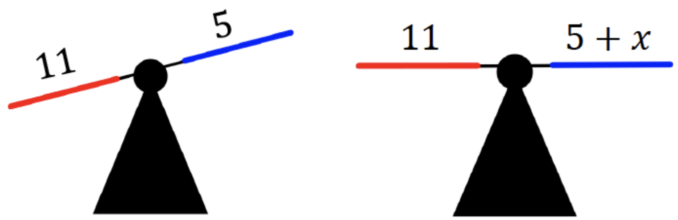
- *What do you notice about the bar model? What are the parts, and what represents the whole?*
- *How does the equation represent the bar model?*

Explore the relationship of inverse operations and equivalent equations. Encourage students to discuss if inverse operations always create equivalent equations. Ask students if they can think of any counterexamples.

12.7.2 Guided Instruction: Solving Equations With Inverse Operations (continued)

d. What is the value of the variable, x?
$x = 3$

3. Two balance scales are shown. The scale on the left is unbalanced. To balance the scale, an unknown value, x, needs to be added to the right side of the scale.



a. The equation represented by the balance scale on the right is shown. Use inverse operations to find the value of x by completing the steps shown.

$x + 5 = 11$

$x + 5 - \boxed{5} = 11 - \boxed{5}$

$x = \boxed{6}$

b. Rewrite the original equation, substituting the solution for x.

$5 + 6 = 11$

c. Explain how to use the equation in Part B to confirm that your solution is correct.
Answers will vary. The equation in Part B confirms that my solution is correct because the value of the expressions on both sides of the equal sign remain equivalent with my solution.

As the class works through question 3, use the following prompts to guide students toward making the connection between the visual of the balanced scales and balancing equations. This will build conceptual understanding that is foundational for lessons moving forward.

- *What do you notice about the balance scales? What is the same? What is different?*
- *How is the second balance scale related to the equation in question 3a?*

12.7.2 Guided Instruction: Solving Equations With Inverse Operations (continued)

Inverse operations can be used to find the value of the variable in an algebraic equation by creating equivalent equations.

Addition Property of Equality

If $a = b$, then $a + c = b + c$

Subtraction Property of Equality

If $a = b$, then $a - c = b - c$

4. The equation $x + 23 = 78$ can be solved by using inverse operations.

   a. Fill in values in each equivalent equation to solve for x.

   $x + 23 = 78$

   $x + 23 - 23 = 78 - 23$

   $x = 55$

   b. What is the inverse operation used to solve the equation $x + 23 = 78$?

   Subtraction

   c. With your partner, discuss how to confirm whether or not the solution is accurate.

   Answers will vary. To confirm that the solution is accurate, I can substitute my solution into the original equation and confirm that the values on each side of the equal sign are equivalent to each other.

MA.K12.MTR.6.1: Reasonableness of Solutions
Refer to MA.K12.MTR.6.1 as you remind students to assess whether their solution is reasonable and to substitute their answer back into the equation as a way to verify their solution.

Have students construct an equivalent equation for $x + 23 = 78$ and explain how and why the equations are equivalent. Also encourage students to construct another pair of equivalent equations using addition or subtraction and then explain how inverse operations can be used to solve them.

12.7.3 Guided Practice: Solving Equations With Inverse Operations

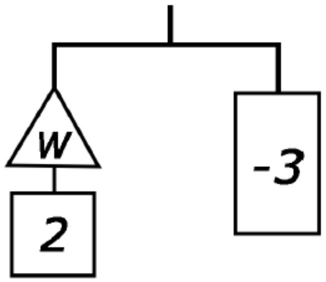
Component Overview: *Students use a hanger, a bar diagram, and a number line to model how to solve one-step equations with inverse operations. Students will continue working with only addition and subtraction. Students may work with a partner to solve the problems together or work independently and then compare their work with a partner's.*

12.7.3 Guided Practice: Solving Equations with Inverse Operations

Work with your partner to complete the following.

An equation can also be represented using a balanced hanger. A hanger stays balanced when the weights on both sides are equal. As the weights are changed, the hanger will stay balanced as long as both sides are changed in the same way.

1. The hanger diagram represents a balanced equation because both sides of the hanger are level.



Discuss that a hanger stays balanced when the weights on both sides are equal. If you change the weights, the hanger will stay balanced as long as both sides are changed in the same way. An equation can be compared to a balanced hanger. You can change the equation, but for a true equation to remain true, the same action must be done on both sides of the equal sign.

   a. Fill in the values for each equation using the hanger diagram. Then, solve for the unknown value, w.

$$\triangle + \boxed{2} = \boxed{-3}$$

$$\triangle + \boxed{2} - \boxed{2} = \boxed{-3} - \boxed{2}$$

$$\triangle = -5$$

Consider modeling this question with an actual hanger that you add and subtract weighted items from, or allow students to work in a small group to create a hanger balance themselves to model the question.

   b. Rewrite the original equation replacing the variable, w, with the solution to check your work.

$-5 + 2 = -3$

12.7.3 Guided Practice: Solving Equations With Inverse Operations (continued)

2. The bar model represents an equation.

x	-4
-9	

   a. Write an addition equation represented by the bar model.
   Answers will vary. $x + (-4) = -9$; $-4 + x = -9$

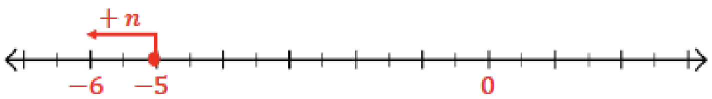
   b. When solving an algebraic equation, the inverse operation can be applied to both sides of the equation. What inverse operation can be used to solve this equation?
   Addition

   c. Write the equivalent equation using inverse operations.
   Answers will vary. $-9 + 4 = x; -9 - x = -4$

   d. What is the value of the variable, x?
   $x = -5$

3. Number line models can be used to represent and solve equations.

   a. Use the number line to represent the solution $-6 = -5 + n$.



   b. What is the value of the variable, n?

   $n = -1$

If students are struggling to create equations, use the following questions to further scaffold:

- *How does the bar model represent the equation?*
- *How can inverse operations be used to solve the equation?*

12.7.4 Your Turn!

Component Overview: *Students will practice using the inverse operations of addition and subtraction to solve one-step equations. Each question uses a different visual model for solving the problem.*

12.7.4 Your Turn!

1. The equation $-31 = s - 6$ can be solved using inverse operations.

   a. Fill in the missing information to solve for s.

   $-31 = s - 6$

   $-31 + 6 = s - 6 + 6$

   $-25 = s$

   b. Which inverse operation was used to solve the equation?

   Addition

   c. Check your solution.

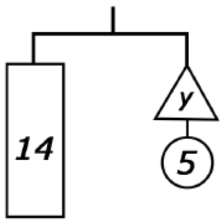
   $-31 = -25 - 6$

   $-31 = -(25 + 6)$

   $-31 = -31$

Solving this question requires students to apply the inverse operation of addition to both sides of the equation. Reiterate the importance of assessing the reasonableness of a solution and substituting the solution back into the equation as a way to check it.

2. The hanger diagram represents an equation.



   a. Write the equation represented by the hanger diagram.

   $14 = y + 5$

   b. What is the value of y?

   $y = 9$

To ensure students are making connections between the representation of the hanger diagram and balancing equations, ask questions like:

- *How does the equation represent the hanger diagram?*
- *How do the different parts of the model relate to the different parts of the equation?*
- *What inverse operation do you use to solve the equation? Why does it work?*
- *How can you check your solution?*

3. Two bar model diagrams are shown.

Bar Model Diagram 1	Bar Model Diagram 2
m	-12
-12 | -4	4 | m

Challenge students to figure out how they could rearrange the numbers and operations to create a different equation. Also have students create other hanger models of equations. If students feel comfortable with the process, allow them to work with the inverse operations of multiplication and division in one-step equations.

a. Write an algebraic equation for each diagram box diagram.

Algebraic Equation for Diagram 1	Algebraic Equation for Diagram 2
Answers will vary. $-12 + (-4) = m$ $m = -4 + (-12)$	*Answers will vary.* $4 + m = -12$ $-12 = m + 4$

b. Find the value of variable, m, for each algebraic equation.

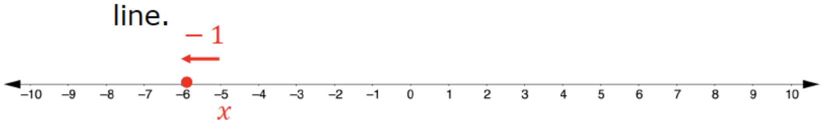
Diagram 1	Diagram 2
$-16 = m$	$m = -16$

c. Discuss with your partner what you notice about the two equations. Summarize your discussion.
Answers will vary. I noticed that that the solutions to both equations are the same even though the equations have different values.

Some students may need a refresher on adding and subtracting integers. If needed, pull a small group of students to review operations with integers.

4. Solve the equation using the number line.

a. Draw a representation for $x - 1 = -6$ on the number line.



b. What is the value of the variable, x?

$x = -5$

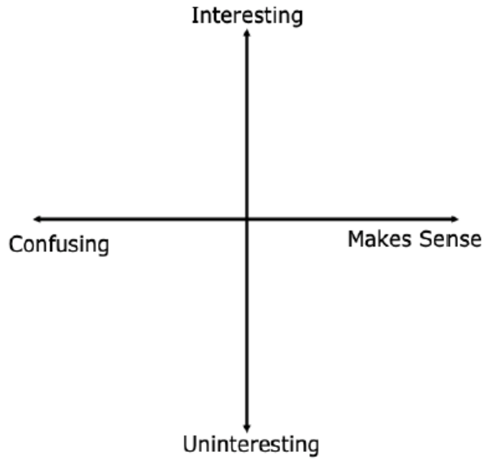
12.7.5 Wrap-Up: Reflection

Component Overview: *Students will reflect on their learning and understanding of the lesson. Encourage students to not only plot a point representing how they feel about meeting the learning target but to also explain why they feel that way.*

12.7.5 Wrap-Up: Reflection

1. Plot a point to indicate how you feel about today's learning target: "I can use diagrams and inverse operations when solving one-step addition and subtraction equations."

   Answers will vary.



Consider prompting students to include the learning target as an ordered pair and explaining why they chose that value. For example, a student may choose $(-1, 5)$ instead of $(-4, 5)$, explaining that it was confusing at first but made more sense by the end of the lesson.